Tangent spaces and derivations

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We will be $\{x^1, ..., x^n\}$ to describe the standard basis of $\mathbb{R}^n$.

Using standard abuse of notation, we will also be using x^i to denote the projection map from $\mathbb{R}^n \to \mathbb{R}$ that extracts the ith coordinate of a given vector.

While initially we will try to maintain generality by giving definitions over manifolds, eventually, we will transition to the specific case of open sets in $\mathbb{R}^n$.

A topological space M is an ***n*-dimensional manifold** iff for every $p \in M$, there is a neighborhood around p which is homeomorphic to some open subset of $\mathbb{R}^n$.

A ***smooth* manifold** is one where we have "diffeomorphic" instead of "homeomorphic".

For every p, the homeomorphism/diffeomorphism $\varphi : U_p \xrightarrow{\sim} U$ from the neighborhood $U_p \subseteq M$ of p to some open subset $U \subseteq \mathbb{R}^n$ is know as a "**chart**".

We will be mostly interested in the case when open subsets of $\mathbb{R}^n$ interpreted as n-manifolds.

Given a smooth n-manifold M, and a point $p \in M$, consider the following set

$$S_p = \left\{ \left(I, \gamma : I \overset{\text{smooth}}{\longrightarrow} M \right) : I \text{ is an open interval in } \mathbb{R} \text{ containing } 0 \text{ s.t. } \gamma(0) = p \right\}$$

So, S_p is the set of smooth curves from some open interval in $\mathbb{R}$ to M "based at" p.

Informally speaking, two smooth curves $(I_1, \gamma_1), (I_2, \gamma_2) \in S_p$ will be considered "equivalent" iff they have the same velocity vector at p.

> To make the above definition formal, first observe that there is a diffeomorphic chart $\varphi : U_p \to U$, where U_p is a neighborhood of p, and U is an open subset in $\mathbb{R}^n$.
>
> We define an equivalence relation $\sim$ on S_p as
>
> $$(I_1, \gamma_1) \sim (I_2, \gamma_2) \Leftrightarrow D(\varphi \circ \gamma_1)(0) = D(\varphi \circ \gamma_2)(0)$$

Finally, we define the **tangent space to M at $p \in M$** as

$$T_p(M) \coloneqq S_p / \sim .$$

Intuitively speaking, $T_p(M)$ stores all possible velocity vectors of smooth curves passing through p.

Equivalently, we can think of members of $T_p(M)$ as the velocity vectors themselves, storing $D(\varphi \circ \gamma)(0)$ instead of $[(I, \gamma)]$.

But we will soon see that the equivalence class definition is nicer in some circumstances.

$[(I, \gamma)] \mapsto D(\varphi \circ \gamma)(0)$ establishes a bijection between $T_p(M)$ and $\mathbb{R}^n$. Observe that this bijection is chart dependent.

The vector space structure on $\mathbb{R}^n$ can be pulled back to establish a vector space structure on $T_p(M)$. So, we get the following vector space isomorphism, for n-dimensional smooth manifolds M,

$$T_p(M) \cong \mathbb{R}^n$$

For the particular case of open subset U or $\mathbb{R}^n$, we get $T_p(U) \cong \mathbb{R}^n \cong T_p(\mathbb{R}^n)$.

Given $f \in C^\infty(M)$ (i.e., f is a C^∞ map from M to $\mathbb{R}$) and $v \in T_p(M)$ (where v is of the form $[(I, \gamma)]$), define the "**directional derivative of f along v**" as

$$D_v(f) := D(f \circ \gamma)(0).$$

When $M = U \overset{\text{open}}{\subseteq} \mathbb{R}^n$, we can apply chain rule to get

$$D_v(f) = D(f \circ \gamma)(0) = Df(\gamma(0)) \times \gamma'(0) = Df(p) \times \gamma'(0)$$

Thinking of $T_p(M)$ as $\mathbb{R}^n$, $\gamma'(0)$ is the data that v carries. We also have that

$$Df(p) = \begin{bmatrix} D_1 f(p) & \dots & D_n f(p) \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x^1}(p) & \dots & \frac{\partial f}{\partial x^n}(p) \end{bmatrix}$$

Therefore, $D_v(f)$ is simply dot product of v with $[D_i f(p)]^T$.

Observe that D_v is an $\mathbb{R}$-linear map from $C^\infty(M)$ to $\mathbb{R}$ which satisfies the **Leibniz property (at p)**,

$$\forall f, g \in C^\infty(M), D_v(fg) = f(p)D_v(g) + D_v(f)g(p)$$

In general, given a smooth n-manifold M, and a point $p \in M$, any $\mathbb{R}$-linear map from $C^\infty(M)$ to $\mathbb{R}$ which satisfies the Leibniz property at p is called a **point derivation at p**.

The set of point derivations at p is denoted as $\text{Der}_p(C^\infty(M), \mathbb{R})$.

Point derivations, as defined above, are intended to capture the essence of "taking the directional derivative at a point". The key idea is that the Leibniz rule is all we care about it anyway.

In fact, as we will soon see, all point derivations at p are "essentially" the same as taking the directional derivative at p along some vector.

Given a topological space X, a set Y, and functions $f, g : X \to Y$, f and g **define the same germ** at $x \in X$ if there is some open neighborhood around x where f and g agree.

Defining the same germ is an equivalence relation on functions of type $X \to Y$.

Given an open set $U \subseteq \mathbb{R}^n$ and a point $p \in U$, consider the set

$$T_{p,U} = \{(V, f) : V \text{ open in } \mathbb{R}^n, p \in V \subseteq U, f \in C^\infty(V)\}.$$

Define the following equivalence relation on T_U,

$$(V, f) \approx (W, g) \Leftrightarrow f|_{V\cap W} \text{ and } g|_{V\cap W} \text{ define the same germ at } p.$$

Define $C^\infty_{p,U}$ as

$$C^\infty_{p,U} := T_{p,U}/\approx .$$

$C^\infty_{p,U}$ forms a vector space over $\mathbb{R}$.

So, each elements of $C^\infty_{p,U}$ contains data about how a C^∞–function can "locally" behave around $p \in U$.

We define point derivations on $C^\infty_{p,U}$ as $\mathbb{R}$-linear maps from $C^\infty_{p,U}$ to $\mathbb{R}$ which satisfy the Leibniz rule at p; i.e., a derivation $\Delta \in \mathrm{Hom}_{\mathbb{R}}\left(C^\infty_{p,U}, \mathbb{R}\right)$ satisfies

$$\forall f, g \in C^\infty_{p,U}, \Delta(fg) = f_0(p)\Delta(g) + \Delta(f)g_0(p),$$
$$\text{where } f = [(V, f_0)], g = [(W, g_0)], fg := [f|_{V\cap W} \times g|_{V\cap W}]$$

The set of **point derivations on $C^\infty_{p,U}$ at p** is denoted as $\mathrm{Der}_p\left(C^\infty_{p,U}, \mathbb{R}\right)$.

In our lectures, $\mathrm{Der}_p\left(C^\infty_{p,U}, \mathbb{R}\right)$ is denoted as $\mathcal{D}\left(C^\infty_p\right)$.

As we mentioned above, each elements in $\mathrm{Der}_p(C^\infty(U), \mathbb{R})$ is supposed to represent the action of taking directional derivative of an arbitrary function at p along some vector.

To extend that analogy, each element in $\mathcal{D}\left(C^\infty_p\right)$ represents taking in the "local behavior at p" of an arbitrary C^∞-function, and then taking its directional derivative along some vector.

It turns out that

$$\mathrm{Der}_p(C^\infty(U), \mathbb{R}) \cong \mathrm{Der}_p\left(C^\infty_{p,U}, \mathbb{R}\right).$$

Try again later. : Get into details.

We will not be dealing with this at all. Handling germs of functions is algebraically nicer if we want to analyze local behavior, so the natural thing to do here is to work in $\mathrm{Der}_p\left(C^\infty_{p,U}, \mathbb{R}\right)$.

This part formalizes that all point derivations at p are actually equal to D_v for some vector v, where D_v represents the taking the directional derivative at p along v.

Given an open set $U \subseteq \mathbb{R}^n$, and a point $p \in U$, we would like to characterize derivations $\Delta \in \mathrm{Der}_p\left(C^\infty_{p,U}, \mathbb{R}\right)$.

Fortunately for us, it turns out that all point derivations are actually equal to D_v for some $v \in T_p(U)$.

In fact,

$$\Phi : T_p(U) \to \mathrm{Der}_p\left(C^\infty_{p,U}, \mathbb{R}\right)$$
$$\Phi := v \mapsto D_v$$

is an isomorphism in $\mathbb{R}$-Vect, and Φ^{-1} is given by

$$\Phi^{-1} : \mathrm{Der}_p\left(C^\infty_{p,U}, \mathbb{R}\right) \to \mathbb{R}^n \cong T_p(U)$$
$$\Phi^{-1} := \Delta \mapsto \left[\Delta\left(\overline{x^1}\right) \ \ldots \ \Delta\left(\overline{x^n}\right)\right]^T,$$

where $\overline{x^i}$ is shorthand for $[(U, x^i|_U)]$, where $x^i : \mathbb{R}^n \to \mathbb{R}$ is the projection map onto the ith basis vector.

Recall the isomorphism between $T_p(U)$ and $\mathbb{R}^n$, $[(I, \gamma)] \mapsto \gamma'(0)$.

Observe that for arbitrary $v \in T_p(U)$, where $v = [(I, \gamma)]$, we have

$$\begin{aligned}
\Phi^{-1}(\Phi(v)) &= \Phi^{-1}(D_v) \\
&= \left[D_v\left(\overline{x^i}\right)\right]^T_{1\le i\le n} \\
&= \left[D_v(x^i)\right]^T_{1\le i\le n} \\
&= \left[D(x^i \circ \gamma)(0)\right]^T_{1\le i\le n} \\
&= \left[D_i(\gamma)(0)\right]^T_{1\le i\le n} \\
&= \gamma'(0),
\end{aligned}$$

and under the isomorphism $\mathbb{R}^n \cong T_p(U)$, this becomes $[(I, \gamma)] = v$. Therefore, $\Phi^{-1} \circ \Phi = \mathrm{id}$.

Now, given an arbitrary point derivation $\Delta \in \mathrm{Der}_p\left(C^\infty_{p,U}, \mathbb{R}\right)$, we have

$$\begin{aligned}
\Phi(\Phi^{-1}(\Delta)) &= \Phi\left(\left[\Delta\left(\overline{x^1}\right) \ \ldots \ \Delta\left(\overline{x^n}\right)\right]^T\right) \\
&= D_v, \text{where } v = \left[\Delta\left(\overline{x^1}\right) \ \ldots \ \Delta\left(\overline{x^n}\right)\right]^T
\end{aligned}$$

We want to show that for this choice of v, $D_v = \Delta$.

We have previously shown that $D_v(f) = \left\langle v, [D_i(f)(p)]^T \right\rangle$, where the inner product is dot product.

So here, for some arbitrary $f \in C^\infty_{p,U}$, we have

$$\begin{aligned}
\Phi(\Phi^{-1}(\Delta))(f) &= D_v(f) \\
&= \sum_{i=1}^n \Delta\left(\overline{x^i}\right) \cdot D_i(f)(p)
\end{aligned}$$

Try again later. : See 22nd Oct board pics, almost done, just do the Taylor expansion.

So, we get that $\mathcal{D}\left(C^\infty_p\right)$ (which is just shorthand for $\mathrm{Der}_p\left(C^\infty_{p,U}, \mathbb{R}\right)$) is exactly the same as $\left\{D_v : v \in T_p(U)\right\}$.

If we think of $T_p(U)$ as being isomorphic to $\mathbb{R}^n$, then we get the basis $\left\{D_{x_i}\right\}_{1\le i\le n}$ for $\mathcal{D}\left(C^\infty_p\right)$. Here $\{x_i\}_{1\le i\le n}$ is the standard basis of $\mathbb{R}^n$.

For example, for $U \overset{\text{open}}{\subseteq} \mathbb{R}^3$, we take the basis $\{e_1, e_2, e_3\}$ for $\mathbb{R}^3 \cong T_p(U)$, which gives us the basis

$$\left\{ D_{e_1} = \left.\frac{\partial}{\partial x}\right|_p, D_{e_2} = \left.\frac{\partial}{\partial y}\right|_p, D_{e_3} = \left.\frac{\partial}{\partial z}\right|_p \right\}$$

for $\mathcal{D}\big(C_p^\infty\big)$.

Therefore we have a 3-way isomorphism so far,

$$T_p(U) \cong \mathbb{R}^n \cong \mathcal{D}\big(C_p^\infty\big),$$

given by

$$v \in \mathbb{R}^n \mapsto \begin{cases} (t \in [-\varepsilon, \varepsilon] \mapsto p + vt) \in T_p(U) \\ (\text{directional derivative along } v) \in \mathcal{D}\big(C_p^\infty\big) \end{cases}$$

$$\gamma \in T_p(U) \mapsto \begin{cases} \varphi'(0) \in \mathbb{R}^n \\ (f \mapsto D(f \circ \gamma)(0)) \in \mathcal{D}\big(C_p^\infty\big) \end{cases}$$

$$\Delta \in \mathcal{D}\big(C_p^\infty\big) \mapsto \begin{cases} \left[\Delta\big(\overline{x^1}\big) \;\; \ldots \;\; \Delta\big(\overline{x^n}\big)\right]^T \in \mathbb{R}^n \\ \text{just use above } \mathbb{R}^n \to T_p(U) \text{ map} \end{cases}$$